Specification-Based Cellular Trust at the Device Edge: A Building Block for the Evolution of Mobile E-Business Information Systems*

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Abstract

Mobile commerce, mobile banking, and Internet of Things (IoT) supply chains rest on the implicit assumption that a User Equipment (UE) is connected to a legitimate cellular network. False Base Station (FBS) attacks break this assumption by exploiting the asymmetric authentication of Long-Term Evolution (LTE) and 5G radio access, exposing e-business platforms to identity theft, transaction fraud, and regulatory exposure under the General Data Protection Regulation (GDPR) and the Network and Information Security Directive 2 (NIS2). This poster paper takes a position-paper view of the radio access trust gap as it affects e-business information systems. It relates the gap to the wider migration of e-business services toward edge-resident architectures, discusses how standards-grounded detection evidence at the device edge may help organizations address audit expectations on data-bearing channels, and sketches practical considerations for enterprises and platform operators that may layer such assurance alongside existing fraud and transaction monitoring stacks. The aim is to surface cellular-layer trust as a topic for the EBISION community rather than to claim a finished solution.

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